

CM3070 Final Project Proposal

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ARIA – Activity Recognition & Inference Architecture, A Multi-Human Activity Detection & Role-Based Tracking System in a Shared Service Counter Environment

Template: CM3065 13.2 – Multi-human Detection and Activities Tracking in a Shared Space

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1. Project Idea

This project proposes the development of ARIA (Activity Recognition & Inference Architecture), a privacy-preserving, multimodal system that detects and tracks the activities of multiple staff members operating in a shared ticketing counter environment. The system identifies who is performing which activity at any given time, using a combination of laptop-based computer vision, ambient audio analysis, and supplementary IoT sensing via an Arduino ESP8266 node.

The target scenario features two distinct staff roles co-existing in one shared physical space: counter agents (stationed at the front, serving customers) and station managers (moving between the front and back office, supervising operations and handling administrative tasks). The system tracks role-specific activities in real time and produces a labelled, publishable dataset as a secondary contribution.

2. Motivation

2.1 The Problem

Shared service environments, such as transport ticketing counters, bank branches, and service desks, routinely involve multiple staff members with different roles performing different activities simultaneously in a single space. Operational decision-making in such environments, such as when to deploy additional counter agents, or when a manager's intervention is needed, currently relies on manual supervision or post-hoc reviews of CCTV footage. Neither approach is efficient, automated, or scalable.

Ambient Assisted Living (AAL) research has made significant progress in detecting activities of daily living, but existing systems are largely designed for single-occupant or undifferentiated multi-occupant settings. They cannot accurately attribute specific activities to specific individuals when multiple people share a space, nor do they address the domain-specific needs of a workplace service counter context.

A further barrier is the privacy and cost implications of cloud-based video processing. Streaming live video to cloud services raises significant data protection concerns, consumes substantial bandwidth, and incurs ongoing financial costs, all identified as open challenges in the brief's background context.

2.2 Why This Project Matters

ARIA addresses this gap by providing:

- Role-aware activity attribution
 - System knows not just what is happening, but who (counter agents vs. station managers) is doing it
- Edge-first inference
 - All vision and ML processing runs locally on a laptop, with only statistical feature vectors transmitted to the cloud
- Privacy by design
 - Cameras are oriented away from customers and no raw video leaves the device
- A novel domain-specific dataset
 - Labelled counter-staff activity data does not currently exist in a published form

3. Related Work

3.1 Nakabayashi & Saito (2024) – Multimodal Human Activity Recognition on Edge Devices

This paper demonstrated that multimodal sensor fusion, combining IMU, depth, and audio on edge hardware, outperforms single-modality approaches for activity recognition in constrained deployment conditions. The authors achieved strong accuracy with a compact model suitable for edge inference. However, their work is limited to single-user scenarios and does not address the challenge of attributing activities to specific individuals in a shared space. Its relevance to ARIA is that it validates the multimodal sensing approach and edge-inference deployment model, highlighting the single-user gap that ARIA targets.

3.2 Li, Gravina, Savaglio & Fortino (2023) – A BSN-Enabled Collaborative Edge-Cloud Architecture for Multi-User Activity Recognition

This paper proposed a hybrid architecture where model training is offloaded to the cloud while inference is performed locally on body sensor networks (BSN). It demonstrated that multi-user recognition is feasible when the architecture separates training in the cloud from inference on edge-devices. However, the system relies on wearable body sensors, which require active participant cooperation and are impractical for a passive service counter environment. Its relevance to ARIA is that it directly validates the hybrid edge-cloud paradigm used in this project. The gap is the wearable dependency, which is why ARIA uses ambient sensing instead.

3.3 Cristina, Despotovic, Pérez-Rodriguez & Aleksic (2024) – Audio- and Video-Based Human Activity Recognition Systems in Healthcare

This survey established that combining audio and video modalities significantly improves human activity recognition (HAR) accuracy over single-modality systems, particularly for distinguishing between activities with similar visual signatures, like standing and sitting. However, the focus is in healthcare settings with individual patients, and multi-person attribution was not addressed. It's relevant to ARIA in that it justifies the audio and vision multimodal approach. The healthcare-to-workplace generalisation gap motivates this project's domain focus.

3.4 Moreaux et al. (2019) – Benchmark for Kitchen20: A Daily Life Dataset for Audio-Based Human Action Recognition

Kitchen20 provides a methodological reference for constructing domain-specific activity datasets using audio signals. The authors documented a structured collection protocol with activity scripting and annotation procedures. However, the dataset is single-person and audio-only, with no vision or IoT sensor fusion. Its relevance to ARIA is that it provides a direct methodological model for ARIA's dataset collection procedure, including activity scripting, labelling protocol, and data card documentation.

3.5 Critical Gap Summary

Across the papers reviewed, no published system:

1. Targets a service counter or workplace role-differentiated scenario
2. Combines passive ambient sensing (no wearables, no customer cameras) with multi-person attribution
3. Produces a publishable annotated dataset specifically for counter-staff activities in a shared space

ARIA looks to fill all three gaps simultaneously.

4. Existing Solutions & Competitive Landscape

Before proposing ARIA, it is necessary to survey what solutions already exist, both in commercial deployments and applied research, to identify where current solutions succeed, where they fall short, and what design decisions ARIA should inherit or avoid.

4.1 Commercial Employee Monitoring Software

Representative systems: DeskTrack, CurrentWare, TrackOlap, Hubstaff, ActivTrak

These platforms monitor staff productivity by tracking screen time, application usage, keystrokes, idle periods, and screenshots on office computers.

Strengths: Mature, production-ready systems with polished dashboards, role-based reporting, and shift-level summaries. Demonstrates what an effective activity monitoring interface looks like in practice.

Limitations: Entirely digital; blind to physical activities such as serving customers, handling documents, walking between zones, or verbal interaction. For counter staff whose core work is physical and interpersonal, these tools offer no relevant insight.

What ARIA learns: Dashboard design; how to present per-role activity summaries, real-time status panels, and shift-level timelines. ARIA's dash interface can borrow this reporting pattern but apply it to physical activities instead of digital ones.

Differentiation: ARIA addresses the physical activity layer these systems are unable to reach.

4.2 Retail Computer Vision & Occupancy Monitoring

Representative systems: Sensormatic, Link Retail, Roboflow People Tracking, ImageVision

These commercial systems use ceiling-mounted cameras and computer vision to count shoppers, measure zone dwell time, and analyse customer movement paths through retail stores.

Strengths: Proven real-time zone-based tracking at scale. Spatial decomposition into named zones (entrance, checkout) is a well-validated design pattern. Some systems report zone occupancy and dwell times in real time.

Limitations: Customer-centric by design; these designs are built to understand shoppers, not staff. They do not distinguish between staff roles, classify what a person is doing (only where they are and how long), and rely on cloud video streaming, raising cost and privacy concerns. Activity classification beyond presence and dwell time is absent.

What ARIA learns: Zone-based spatial decomposition maps directly to ARIA's Counter Zone A and Back Zone B. Dwell time per zone per tracked individual becomes a useful classifier feature. The camera-mounting angle, overhead or behind-counter, is also validated by this industry.

Differentiation: ARIA classifies what staff are doing, not just where they are. It attributes activities to named roles, runs inference locally, and transmits no raw video to the cloud.

4.3 Ambient Assisted Living (AAL) Research Systems

Representative works: Multi-view HAR survey (2025), AAL for frail persons (Frontiers, 2023), multi-modal deep learning HAR (arXiv, 2026)

Academic AAL systems deploy sensors, like wearables, passive infrared, cameras, and microphones, in home environments to detect activities of daily living, like sitting, walking, cooking, or falling, for elderly or vulnerable individuals.

Strengths: Richly validated sensor fusion methodologies (data-level, feature-level, decision-level). Multi-modal approaches consistently outperform single-modality baselines. A strong evaluation culture: per-class F1, confusion matrices, leave-one-subject-out cross-validation are standard practices.

Limitations: Designed for single-occupant domestic environments. Multi-person attribution in shared spaces is an explicitly acknowledged open problem in the literature. Many systems require wearable sensors, which assume continuous participant cooperation and are unsuitable for a naturalistic service counter setting.

What ARIA learns: Feature-level sensor fusion architecture; combining vision keypoints, audio amplitudes, and IoT sensor readings into a unified feature vector before classification. Evaluation protocols; per-class F1, confusion matrix, leave-one-session-out splits. Activity taxonomy design: defining mutually exclusive, exhaustive activity classes with clear discriminative signatures.

Differentiation: ARIA operates in a multi-occupant workplace setting with two distinct staff roles, the exact scenario AAL literature identifies as an unsolved extension of their domestic single-user work.

4.4 Computer Vision for Workspace Labour Tracking

Representative works: CV-based worker motion estimation in manufacturing (Leverage, 2025)

Emerging industrial CV deployments use overhead cameras to estimate worker motion amounts in factories and warehouses, quantifying how much physical movement a worker performs in a shift.

Strengths: Establishes feasibility of vision-based physical labour analysis in professional settings. Demonstrates that skeletal keypoint velocity is a meaningful proxy for activity intensity.

Limitations: Manufacturing context only; repetitive, single-worker, single-task scenarios. No role differentiation between worker types. No multimodal fusion (audio, IoT). No activity type classification, only motion quality.

What ARIA learns: Skeleton joint velocity (derived from MediaPipe/YOLOv8-pose keypoints across frames) as a feature in the classifier, distinguishing high-motion activities (walking, reaching) from low-motion ones (typing, on the phone). This supplements visual appearance and zone-sensor signals.

Differentiation: ARIA classifies what kind of activity is occurring, not just how much movement. It distinguishes between eight role-specific activity types rather than a single motion quantity metric.

4.5 Synthesis: Why ARIA is Needed

Table 1: ARIA Synthesis

System Category	Role Attribution	Physical Activity Classification	Multimodal Fusion	Edge Inference	Service Counter Domain	Open Dataset
Commercial monitoring software	✗	✗	✗	✓	✗	✗
Retail CV / occupancy	✗	✗	✗	✗	Partial	✗
AAL research systems	✗	✓	✓	Partial	✗	Partial

Workplace labour CV	✗	Partial	✗	✗	✗	✗
ARIA	✓	✓	✓	✓	✓	✓

No existing system satisfies all six criteria simultaneously. ARIA is positioned at the intersection of these four solution categories, synthesising their strengths while addressing shared gaps.

5. Project Concept

5.1 System Overview

ARIA operates across the following three tiers:

5.1.1 Tier 1 – Edge Sensor Node (Arduino ESP8266)

Three PIR motion sensors and two ultrasonic distance sensors are deployed across two named zones (counter zone, back zone), with an analogue microphone module on A0. The ESP8266 reads all sensors and transmits JSON payloads over serial USB to the edge inference host.

5.1.2 Tier 2 – Edge Inference Host (Laptop/PC)

A Python application subscribes to the ESP8266 serial stream and simultaneously captures webcam video and microphone audio. A YOLOv8n model detects and tracks persons (ByteTrack); a pose estimation model (MediaPipe or YOLOv8-pose) extracts skeletal keypoints. Combined with sensor features, a 1D-CNN or Random Forest classifier identifies the activity class of each tracked individual. Results are logged to a timestamped CSV dataset and displayed on a real-time Dash dashboard.

5.1.3 Tier 3 – Cloud Layer (Google Colab + Firebase)

Feature vectors, not raw video, are periodically synced to Firebase Realtime Database. Google Colab retrains the classifier on accumulated data and pushes updated model weights back to the inference host in an iterative loop.

5.2 Activity Classes

Table 2: Activity Classes

Role	Activity Classes
Counter Agent	Serving (facing forward, leaning in), typing/processing (keyboard interaction), idle/waiting, reaching/handling
Station Manager	Walking (zone transition), desk work (seated, screen/paperwork), in-conversation (facing another staff member)

5.3 Hardware

- Laptop/PC – edge inference host (existing hardware)
- Arduino ESP8266 NodeMCU – IoT Sensor Hub
- 3 x HC-SR501 PIR Sensor – motion detection per zone
- 2 x HC-SR04 Ultrasonic Sensor – proximity/zone confirmation
- MAX9814 Microphone Module – ambient audio amplitude

- USB Webcam

6. Ethical Considerations

This project involves recording human participants in a simulated counter environment. The following measures will be implemented:

- Participant profile: Healthy adult volunteers only. No children, vulnerable adults, medical patients, or minors will participate.
- Informed consent: All participants will receive a written Participant Information Sheet prior to involvement and sign a written consent form. The form will cover the purpose of the study, what data is collected, how it is stored, the right to withdraw at any time without consequence, and how long the data will be retained.
- Simulated environment: Scenario will be replicated in a controlled room setting with volunteer participants roleplaying the counter agent and manager roles. No real customers will be present or filmed.
- Camera orientation: Camera is positioned behind the staff area, oriented away from any public-facing direction.
- ROI masking: Software-level Region of Interest masking is applied before any frame is processed or stored, ensuring only the designated staff zone is captured.
- Data minimisation: No raw video is stored beyond the local session. Only feature vectors and activity labels are retained for analysis and dataset publication.
- UK GDPR compliance: All data handling follows the UK Data Protection Act 2018 / UK GDPR principles: lawfulness, fairness, transparency, purpose limitation, data minimisation, storage limitation, and integrity.
- Dataset publication: Raw participant-identifiable data will not be published. The dataset will contain only anonymised sensor readings with activity labels. Participants will provide explicit consent for their anonymised data to be included in any published dataset.

7. Proposed Outcomes

Minimum (Pass):

A prototype running on a PC/edge device that can process vision and audio signals to detect and track at least two activity types across two simultaneous participants.

Good (Merit):

A hybrid edge-cloud prototype demonstrating the full training-inference-retraining loop, accurately detecting more than or at least 4 activity types. A sufficiently large annotated dataset with documentation.

Outstanding (1st):

As above, plus a robustly validated system with per-class F1 metrics, ablation study results, and a publishable labelled dataset with full data card documentation, produced with the written consent of participants and suitable for submission to a public data repository like Kaggle or Zenodo.

8. References

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